STANDATA interpretation 23-BCI-011/24-ECI-009/23-FCI-006

Building/Electrical/Fire

Existing fire alarm systems

Date Issued: March 2025

Purpose

To clarify requirements where an existing fire alarm system (FAS), regardless of age, requires inspection, testing, maintenance, and installation of devices or other equipment (e.g., maglocks) to provide or maintain an expected level of safety for occupants.

In addition, this interpretation defines who can install, verify, inspect, test, and maintain an FAS.

Discussion

There are five types of activities conducted on an FAS: installation, verification, inspection, testing, and maintenance.

1. Installation includes a new install or additions, modifications, and renovations (changes) to an existing installation.

2. Verification involves inspection and testing for verifying the FAS conforms to the design, and applicable codes and standards, and performs all its intended functions.

3. Inspection is a visual examination to determine the device or system will apparently perform in accordance with its intended function.

4. Testing is the operation of a device or system to determine it will perform in accordance with its intended operation or function.

5. Maintenance is the removal, replacement, or servicing of devices or equipment found inoperative during an inspection and testing of the FAS. This would include the discovery of an inoperative device at any other time outside of the scheduled mandatory inspections.

The assessment of the existing FAS design may impact the extent of work required to maintain safety standards. This includes systems that were installed following "good engineering practices" prior to the adoption of current codes and standards.

It also highlights the importance of considering the FAS installed in accordance with established codes and standards when planning maintenance. It is essential to determine the extent to which the requirements of the most current codes and standards apply to the maintenance and modifications of these systems.

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The main codes containing specific requirements are the National Building Code – 2023 Alberta Edition (NBC(AE)), the Canadian Electrical Code (CE Code) 2024, and the National Fire Code – 2023 Alberta Edition (NFC(AE)). These codes reference the following standards for the installation, verification, inspection, and testing of an FAS:

- CAN/ULC-S524, “Standard for Installation of Fire Alarm Systems”
- CAN/ULC-S536, “Standard for Inspection and Testing of Fire Alarm Systems”
- CAN/ULC-S537, “Standard for Verification of Fire Alarm Systems”
- CAN/ULC-S1001, “Standard for the Integrated Systems Testing of Fire Protection and Life Safety Systems”

Installations, verifications, inspections, testing, and maintenance of fire alarm and voice communication systems can only be completed by those recognized as qualified to do so under Division C of the NFC(AE).

Safety Codes Act

Interpretation
1(1) In this Act,
(e) “Administrator” means an Administrator appointed under this Act;

Application of Act
2(1) This Act applies to fire protection, barrier-free design and the design, manufacture, construction, installation, use, operation, occupancy and maintenance of
a) buildings,
b) electrical systems,
c) elevating devices,
d) gas systems,
e) plumbing systems,
f) pressure equipment, and
g) private sewage disposal systems.

Permit Regulation

Electrical permit
8(1) A permit in the electrical discipline is required to install, alter or add to an electrical system.

Code References

NBC(AE) and NFC(AE) Preface

Relationship between the NBC(AE) and the NFC(AE)
The NBC(AE) and NFC(AE) each contain provisions that relate to the safety of persons in buildings in the event of a fire and the protection of buildings from the effects of fire.(1) These two codes are developed as complementary and coordinated documents to minimize the possibility of their containing conflicting provisions. It is expected that buildings comply with both the NBC(AE) and the NFC(AE).

The NBC(AE) covers the fire safety and fire protection features that are required to be incorporated in a building at the time of its original construction. Building codes typically no longer apply once a building is occupied, unless the building is undergoing alteration or change of use, or being demolished.

The NFC(AE) covers:

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- the ongoing maintenance and use of the fire safety and fire protection features incorporated in buildings
- the conduct of activities that might cause fire hazards in and around buildings
- limitations on hazardous contents in and around buildings
- the establishment of fire safety plans
- fire safety at construction and demolition sites

In addition, the NFC(AE) contains provisions regarding fire safety and fire protection features that must be added to existing buildings when certain hazardous activities or processes are introduced into these buildings.

Some of the NFC(AE)'s provisions are incorporated by reference in the NBC(AE) and, thus, may apply to original construction, alterations, or changes in use.

NBC(AE) or NFC(AE), Division A, Article 1.4.1.2.

1.4.1.2. Defined Terms
1) The words and terms in italics in this Code shall have the following meanings
Approved means acceptable to the *Provincial Fire Administrator*.

Authority having jurisdiction means a safety codes officer in the fire discipline exercising authority pursuant to designation of powers and terms of employment in accordance with the Safety Codes Act.

Provincial Building Administrator means a person in the building discipline appointed as an Administrator under the Safety Codes Act.

Provincial Fire Administrator means a person in the fire discipline appointed as an Administrator under the Safety Codes Act.

Registered engineering professional means an individual who is authorized to engage in the practice of engineering under the Engineering and Geoscience Professions Act and its Regulations.

A-1.4.1.2.(1) Defined Terms.
Authority Having Jurisdiction
The defined term “authority having jurisdiction” refers to a safety codes officer as the appointed individual responsible for decision-making in a number of areas. In most cases, this individual will be a member of an accredited municipal fire department. In municipalities that use accredited agencies for their inspection and enforcement, the fire department may not be involved. It is important that accredited municipalities employing accredited agencies properly designate the person(s) acting as the authority having jurisdiction.

In addition, municipalities using accredited agencies should ensure that the agency has a close working relationship with the fire department during the decision-making process.

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NBC(AE), Division A, Article 1.1.1.2.

1.1.1.2. Application to Existing Buildings (See Note A-1.1.1.2.)
1) This Article applies to a *building* that legally existed before 01 May 2024.
2) If a *building* is altered, rehabilitated, refurbished, renovated or repaired, the level of life safety and *building* performance shall not be decreased.
3) Except as specified in Part 10 of Division B, the *authority having jurisdiction* shall accept any construction or condition that lawfully exists in Alberta.
4) A change in *occupancy* or *alteration* of any *building* shall be permitted if the level of safety and *building* performance proposed are acceptable to the *authority having jurisdiction*.
5) The *authority having jurisdiction* may accept an alternative or a proposal that achieves the appropriate level of safety for the specific activity for which the *building* is to be used.
6) The *authority having jurisdiction* may accept existing construction not in complete compliance with this Code and may place conditions on the existing construction.

A-1.1.1.2. Application to Existing Buildings. This Code is most often applied to existing or relocated buildings when an owner wishes to rehabilitate a building, change its use, or build an addition, or when an enforcement authority decrees that a building or class of buildings be altered for reasons of public safety. It is not intended that the NBC(AE) be used to enforce the retrospective application of new requirements to existing buildings or existing portions of relocated buildings. For example, although the NFC(AE) could be interpreted to require the installation of fire alarm, standpipe and hose, and automatic sprinkler systems in an existing building for which there were no requirements at the time of construction, it is the intent of Alberta Municipal Affairs that the NFC(AE) not be applied in this manner to these buildings unless the authority having jurisdiction has determined that there is an inherent threat to occupant safety and has issued an order to eliminate the unsafe condition, or where substantial changes or additions are being made to an existing building or the occupancy has been changed. (See also Note A-1.1.1.1.(1) of Division A of the NFC(AE).)

Relocated buildings that have been in use in another location for a number of years can be considered as existing buildings, in part, and the same analytical process can be applied as for existing buildings. It should be noted, however, that a change in occupancy may affect some requirements (e.g. loads and fire separations) and relocation to an area with different wind, snow or earthquake loads will require the application of current code requirements. Depending on the construction of the building and the changes in load, structural modifications may be required. Similarly, parts of a relocated or existing building that are reconstructed, such as foundations and basements, or parts being modified are required to be built to current codes.

Whatever the reason, Code application to existing or relocated buildings requires careful consideration of the level of safety needed for that building. This consideration involves an analytical process similar to that required to assess alternative design proposals for new construction. See Clause 1.2.1.1.(1)(b) for information on achieving compliance with the Code using alternative solutions.

In developing Code requirements for new buildings, consideration has been given to the cost they impose on a design in relation to the perceived benefits in terms of safety. The former is definable; the latter difficult to establish on a quantitative basis. In applying the Code requirements to an existing building, the benefits derived are the same as in new buildings. On the other hand, the increased cost of implementing in an existing building a design solution that would normally be intended for a new building may be prohibitive.

The successful application of Code requirements to existing construction becomes a matter of balancing the cost of implementing a requirement with the relative importance of that requirement to the overall Code objectives. The degree to which any particular requirement can be relaxed without affecting the

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intended level of safety of the Code requires considerable judgment on the part of both the designer and the authority having jurisdiction.

Further information on the application of Code requirements to existing or relocated buildings can be found in the following publications:

- "User's Guide – NBC 1995 Fire Protection, Occupant Safety and Accessibility (Part 3)"
- "Guidelines for Application of Part 3 of the National Building Code of Canada to Existing Buildings"
- Commentary entitled "Application of NBC Part 4 of Division B for the Structural Evaluation and Upgrading of Existing Buildings" of the "Structural Commentaries (User's Guide – NBC 2020: Part 4 of Division B)"
- "User's Guide – NBC 1995, Application of Part 9 to Existing Buildings"
- CBD 230, "Applying building codes to existing buildings"

These publications can be accessed through the NRC Publications Archive at NRC Publications Archive - Canada.ca

NBC(AE), Division A, Article 3.2.4.5.

3.2.4.5. Installation and Verification of Fire Alarm Systems
1) Except as permitted by Article 3.2.4.19., fire alarm systems, including the voice communication capability where provided, shall be installed in conformance with CAN/ULC-S524, "Standard for Installation of Fire Alarm Systems."
2) Fire alarm systems shall be verified in conformance with CAN/ULC-S537, "Standard for Verification of Fire Alarm Systems," to ensure they are operating satisfactorily.

NBC(AE), Division C, Article 2.4.1.4.

2.4.1.4. Fire Alarm System Designs
1) If a fire alarm system is to be installed in a *building* described in Sentence 2.4.1.1.(3), the owner shall submit evidence to the *authority having jurisdiction*, before construction begins, that they have retained a *registered professional* to
a) design the system,
b) perform *field reviews* of the system during installation, and
c) witness verification of the system after installation.

NFC(AE), Division A, Articles 1.1.1.1. and 1.2.1.1.

1.1.1.1. Application of this Code
1) This Code applies to all new and existing *buildings* and facilities, and to *building* construction, renovation or demolition sites. (See Note A-1.1.1.1.(1).)

A-1.1.1.1.(1) Application of this Code. This Code applies to buildings and facilities, whether occupied or vacant. For the purposes of Sentence 1.1.1.1.(1), the term "facilities" is used in its broadest sense to include all premises that are not included in the definition of "building" in this Code, such as outdoor and underground areas, structures and equipment. Such "facilities" are often associated with storage, distribution and manufacturing activities.

The NFC(AE) contains references to the National Building Code – 2023 Alberta Edition (NBC(AE)) for the

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design, construction and installation of many fire protection features. Some NBC(AE) requirements are most readily applied to new buildings and their retroactive application to existing situations as prescribed by this Code could result in some difficulty in achieving compliance. It is the intent of the NFC(AE) that an equivalent level of safety be achieved rather than necessarily achieving strict conformance to the NBC(AE). The application of this Code to the upgrading of existing facilities should be based on the judgment of the enforcement authority, who must deal with each case on its own merits.

The NFC(AE) states that the owner or the owner's authorized agent is responsible for carrying out the provisions of the Code (see Article 2.2.1.1. of Division C). However, the owner is expected to communicate with the authority having jurisdiction, who is in a position to assess the relative significance of variances from the NBC(AE) requirements. Such authority may then determine that upgrading measures are not necessary, on the basis that the existing arrangement represents an equivalent level of fire and life safety. The NFC(AE) presumes that the adopting legislation provides for the exercise of the necessary discretionary judgment on the part of the enforcing officials, along with appropriate rights to appeal (see Note A-2.2. of Division C). See also Note A-2.1.3.1.(1) of Division B and Note A-1.1.1.1.(1) of Division A of the NBC(AE).

1.2.1.1. Compliance with this Code

1) Compliance with this Code shall be achieved by

a) complying with the applicable acceptable solutions in Division B (see Note A-1.2.1.1.(1)(a)), or

b) using alternative solutions that will achieve at least the minimum level of performance required by Division B in the areas defined by the objectives and functional statements attributed to the applicable acceptable solutions (see Note A-1.2.1.1.(1)(b)).

A-1.2.1.1.(1)(a) Code Compliance via Acceptable Solutions. If a building design (e.g. material, component, assembly or system) can be shown to meet all provisions of the applicable acceptable solutions in Division B (e.g. it complies with the applicable provisions of a referenced standard), it is deemed to have satisfied the objectives and functional statements linked to those provisions and thus to have complied with that part of the Code. In fact, if it can be determined that a design meets all the applicable acceptable solutions in Division B, there is no need to consult the objectives and functional statements in Division A to determine its compliance.

A-1.2.1.1.(1)(b) Code Compliance via Alternative Solutions. Where a design differs from the acceptable solutions in Division B, then it should be treated as an “alternative solution.” A proponent of an alternative solution must demonstrate that the alternative solution addresses the same issues as the applicable acceptable solutions in Division B and their attributed objectives and functional statements. However, because the objectives and functional statements are entirely qualitative, demonstrating compliance with them in isolation is not possible. Therefore, Clause 1.2.1.1.(1)(b) identifies the principle that Division B establishes the quantitative performance targets that alternative solutions must meet. In many cases, these targets are not defined very precisely by the acceptable solutions—certainly far less precisely than would be the case with a true performance code, which would have quantitative performance targets and prescribed methods of performance measurement for all aspects of building performance. Nevertheless, Clause 1.2.1.1.(1)(b) makes it clear that an effort must be made to demonstrate that an alternative solution will perform as well as a design that would satisfy the applicable acceptable solutions in Division B—not “well enough” but “as well as.”
In this sense, it is Division B that defines the boundaries between acceptable risks and the “unacceptable” risks referred to in the statements of the Code's objectives, i.e. the risk remaining once the applicable acceptable solutions in Division B have been implemented represents the residual level of risk deemed to be acceptable by

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the broad base of Canadians who have taken part in the consensus process used to develop the Code.

Level of Performance

Where Division B offers a choice between several possible designs, it is likely that these designs may not all provide exactly the same level of performance. Among a number of possible designs satisfying acceptable solutions in Division B, the design providing the lowest level of performance should generally be considered to establish the minimum acceptable level of performance to be used in evaluating alternative solutions for compliance with the Code.

Sometimes a single design will be used as an alternative solution to several sets of acceptable solutions in Division B. In this case, the level of performance required of the alternative solution should be at least equivalent to the overall level of performance established by all the applicable sets of acceptable solutions taken as a whole.

Each provision in Division B has been analyzed to determine what it is intended to achieve. The resultant intent statements clarify what undesirable results each provision seeks to preclude. These statements are not a legal component of the Code, but are advisory in nature, and can help Code users establish performance targets for alternative solutions. They are published as part of the online Code subscriptions and as a separate electronic document entitled “Supplement to the NFC 2015: Intent Statements,” which is available on NRC's Web site. It is important to note that intent statements are not available for requirements that are unique to the NFC(AE).

Areas of Performance

A subset of the acceptable solutions in Division B may establish criteria for particular types of designs (e.g. certain types of materials, components, assemblies, or systems). Often such subsets of acceptable solutions are all attributed to the same objective: Fire Safety for example. In some cases, the designs that are normally used to satisfy this subset of acceptable solutions might also provide some benefits that could be related to some other objective: Fire Protection of the Building or Facility for example. However, if none of the applicable acceptable solutions are linked to Objective OP1, Fire Protection of the Building or Facility, it is not necessary that alternative solutions proposed to replace these acceptable solutions provide a similar benefit related to Fire Protection of the Building or Facility. In other words, the acceptable solutions in Division B establish acceptable levels of performance for compliance with the Code only in those areas defined by the objectives and functional statements attributed to the acceptable solutions.

Applicable Acceptable Solutions

In demonstrating that an alternative solution will perform as well as a design that would satisfy the applicable acceptable solutions in Division B, its evaluation should not be limited to comparison with the acceptable solutions to which an alternative is proposed. It is possible that acceptable solutions elsewhere in the Code also apply. The proposed alternative solution may be shown to perform as well as the most apparent acceptable solution, which it is replacing, but may not perform as well as other relevant acceptable solutions. For example, the use of sprinklers to protect the exterior wall of a building may permit combustible materials to be stored closer to that wall than otherwise permitted by the Code, but the proximity of the stored materials to the wall may contravene firefighter access provisions elsewhere in the Code. All applicable acceptable solutions should be taken into consideration in demonstrating the compliance of an alternative solution.

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NFC(AE), Division A, Article 2.1.1.1.

2.1.1.1. Application
1) This Part applies to all *buildings* and facilities in this Code. (See Article 1.1.1.1)

NFC(AE), Division B, Article 6.1.1.1

6.1.1.1. Application
1) This Part includes requirements for the inspection, testing, maintenance, and operation of portable extinguishers, water-based fire protection systems, special extinguishing systems, fire alarm systems, emergency electrical power supply systems, emergency lighting, and exit signs.

NFC(AE), Division B, Articles 6.3.1.2. and 6.3.1.4.

6.3.1.2. Inspection and Testing
1) Fire alarm systems shall be inspected and tested in conformance with CAN/ULC-S536, “Standard for Inspection and Testing of Fire Alarm Systems.”

6.3.1.4. Voice Communication Systems
1) Voice communication systems that are integrated with a required fire alarm system shall be tested in conformance with Article 6.3.1.2.

NFC(AE), Division C, Article 2.2.4.3.

2.2.4.3. Fire Alarm and Voice Communication Systems
1) Persons are considered qualified in the installation of fire alarm and voice communication systems when they have acquired
- a) an Alberta journeyperson's electrician certificate on or after September 1, 1991,
- b) an Alberta journeyperson's electrician certificate prior to September 1, 1991, and have completed a fire alarm course recognized by the *Provincial Fire Administrator*, or
- c) a Canadian Red Seal journeyperson's electrician certificate and have completed a fire alarm course recognized by the *Provincial Fire Administrator*. (See Note A-2.2.4.3.(1).)

2) Persons are considered qualified to make operational, inspect, test and maintain fire alarm and voice communication systems and sign applicable documentation when they have acquired
- a) a certificate of training in this area of study from a public post-secondary educational institution, including
  - i) an Alberta journeyperson's electrician certificate on or after September 1, 1991,
  - ii) an Alberta journeyperson's electrician certificate prior to September 1, 1991, and a fire alarm course recognized by the *Provincial Fire Administrator*, or
  - iii) a Canadian Red Seal journeyperson's electrician certificate and a fire alarm course recognized by the *Provincial Fire Administrator*, or
- b) Fire Alarm Technician certification issued by the Canadian Fire Alarm Association (CFAA) and that certification shall be maintained as per the CFAA's certification policy.

3) An apprentice electrician may complete all the work a journeyperson electrician is able to, provided
- a) the apprentice is competent to complete the work, and
- b) the apprentice is under the supervision of a journeyperson electrician in accordance with the Designated Trades and Restricted Activities Regulation.

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4) A CFAA fire alarm student or trainee may complete all of the work a certified technician is able to, provided

a) the CFAA fire alarm student or trainee is actively enrolled in the CFAA Fire Alarm Technician program, and

b) the CFAA fire alarm student or trainee is competent to complete the work and is supervised on site by a certified CFAA technician. (See Note A-2.2.4.3.(4).)

5) A person who assists with the verification, annual inspection, testing and maintenance of a fire alarm system at the control panel shall meet the qualifications outlined in Sentences (1) to (4).

6) The daily and monthly inspections and tests required by CAN/ULC-S536, "Standard for Inspection and Testing of Fire Alarm Systems," may be completed by

a) the owner, or

b) the owner's representative.

7) The owner and owner's representative referred to in Sentence (6) shall receive training by the manufacturer or service provider on the required procedures for their specific fire alarm system.

A-2.2.4.3.(1) The types of training provided and other provincial legislation may limit the scope of activities a qualified person may perform on such systems.

Fire alarm system verifications are generally the responsibility of a licensed engineering professional. In Alberta, this must be the registered professional who will complete a Schedule C under the NBC(AE).

The NFC(AE) states that a person who has acquired an approved certificate of training from a post-secondary educational institution may also inspect, test and maintain fire alarm and voice communication systems. Prior to the NFC(AE), Division C Appendix (A-2.2.4.3.(1)) stated a "Fire Alarm Technician certification issued by the Canadian Fire Alarm Association (CFAA) or the Northern & Southern Alberta Institutes of Technology." This information was not included in the current Notes to Division C for the NFC(AE).

Those persons who have Fire Alarm Technician certification from the Northern Alberta Institute of Technology or Southern Alberta Institute of Technology are still qualified to inspect, test and maintain fire alarm and voice communication systems until December 31, 2025. To continue to inspect, test and maintain these systems after this date, persons must obtain a certificate of training from the CFAA.
Prior to 2000, Alberta Labour issued a "P" ticket number or a personal identification number (PIN) card to certified electricians, manufacturer-trained technicians, electronics technicians, electrical engineering technologists, and professional electrical engineers. This P card was to confirm that the individual had successfully passed their training to conduct work on fire alarm systems.

In 2000, the practice of issuing a P ticket or PIN was discontinued. As Alberta Municipal Affairs no longer has the Alberta Labour program in place, the confirmation of a P ticket or PIN can no longer be provided. Anyone still in possession of their laminated P card should note that the card is still valid until December 31, 2025.

Cards issued by the Alberta Fire Safety Association (AFSA) will no longer be recognized as proof of certification effective December 31, 2025.

Electrical Contractors Association of Alberta (ECAA)

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To obtain proof of certification in lieu of the above-mentioned cards, individuals must contact the Electrical Contractors Association of Alberta (ECAA). There are two options to obtain CFAA designation:

Option 1: Individuals may take a 40-hour upgrade course offered by ECAA, followed by the CFAA theory and practical exam for CFAA registration.

Option 2: Individuals may take the CFAA's Courses 1 and 5 (for equivalent courses, contact the ECAA), followed by the CFAA theory and practical exam for CFAA registration.

The ECAA has implemented a new PIN program. Journeypersons who have lost, misplaced or had their Alberta Labour P ticket or PIN card stolen will need to work with the ECAA to obtain a PIN card if they still wish to carry this card.

A-2.2.4.3.(4) A CFAA fire alarm student or trainee is someone who is actively enrolled in the CFAA Fire Alarm Technology Education program.

NFC(AE), Division C, Sentence 2.2.9.2(1)

2.2.9.2. Fire Alarm and Voice Communication System Audits

1) When an existing fire alarm system has not been subjected to an initial verification as required by CAN/ULC-S524, “Standard for Installation of Fire Alarm Systems,” and described in CAN/ULC-S537, “Standard for the Verification of Fire Alarm Systems,” the authority having jurisdiction may require the system to be audited in conformance with CAN/ULC-S536, “Standard for Inspection and Testing of Fire Alarm Systems,” unless the authority having jurisdiction is satisfied that the system has been maintained and is functioning in a manner acceptable to the authority having jurisdiction.

C22.1:24 – 2024 Canadian Electrical Code, Part 1

32-000 Scope (see Appendix B)

1) This Section applies to the installation of electrical local fire alarm systems, permanently connected smoke alarms, carbon monoxide alarms, and fire pumps required by the National Building Code of Canada.

C22.1:24 – 2024 Canadian Electrical Code, Part 1 Appendix B

Rule 32-000

For further information pertaining to the installation of fire alarm systems, reference should be made to CAN/ULC-S524.

It is essential that fire alarm systems be maintained in an operating condition at all times. The inspection, maintenance, and testing procedures are detailed in the *National Fire Code of Canada* or the appropriate provincial fire code.

For further information pertaining to the installation of fire pumps, reference should be made to NFPA 20.

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Application

This interpretation applies to all buildings where an FAS is currently in existence and the fire alarm undergoes installation, verification, inspection, testing, or maintenance.

Interpretation

1. Only those qualified as per the NFC(AE) can complete these types of activities.

2. Qualified electricians and fire alarm technicians can complete maintenance activities as per NFC(AE) Division C, but only qualified electricians can replace or repair wiring.

3. For an FAS installed under previous Alberta editions of the building code, the replacement of devices with equivalent devices is permitted (like-for-like device replacement). However, if the replacement is not an equal exchange, the following scenarios are to be followed:

- Change of use or modernization of the building: the current NBC(AE) applies.
- No change of use: in this situation, upgrading to the current NBC(AE) could be onerous and according to the NFC(AE) Division A, Article 1.1.1.1., the owner may apply to the authority having jurisdiction (AHJ) (of the NBC(AE)) for an alternative solution detailing the reasons why they need a fire alarm replaced and why they want to install a new fire alarm to an older edition of the Alberta Building Code and/or CAN/ULC standard.

NOTE: Where the NBC(AE) requires professional involvement for a building due to its size, occupancy, or complexity, a registered engineering professional is also required to witness verification of upgrades or modifications on an existing FAS.

If an FAS requires additions or modifications, these changes are still categorized as installs and, along with completely new installs, can only be completed by a qualified electrician.

The following chart shows the work which an electrician or fire alarm technician can perform:

Work Acceptable Under the Designated Trades and Restricted Activities Regulation and the NFC(AE)

INSTALLATION & MAINTENANCE OF FIRE ALARM SYSTEMS		
	ELECTRICIAN	**TECHNICIAN**
Installation	✓	✗
Inspection	✓	✓
Testing	✓	✓
Maintenance	✓ (1)	✓ (1)
Additions / Renovations	✓	✗
Audit System (Division C, 2.2.9.2.)	✗ (2)	✗ (2)

Notes to Table:

(1) Electricians and fire alarm technicians must be qualified under the NFC(AE).

(2) Electricians and fire alarm technicians qualified as per the NFC(AE) may perform an audit where professional involvement is not required as per the NBC(AE).

The Permit Regulation made pursuant to the *Safety Codes Act* outlines the necessary permit requirements for both building and electrical disciplines.

It is essential to contact the local AHJ prior to initiating any work on the FAS.

Fire alarm system verification to be completed in accordance with CAN/ULC S537, "Verification of Fire Alarm Systems."

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This INTERPRETATION is applicable throughout the Province of Alberta.

Unless stated otherwise, all Code references in this STANDATA are from the National Building Code –- 2023 Alberta Edition, National, Fire Code – 2023 Alberta Edition and the Canadian Electrical Code

Issued by the Provincial Building, Electrical and Fire Administrator

[Original Signed]
Paul Chang
Provincial Building Administrator

[Original Signed]
Tina Parker
Provincial Fire Administrator

[Original Signed]
Kevin Glubrecht
Provincial Electrical Administrator

Alberta Municipal Affairs – Technical and Corporate Services
Email: safety.services@gov.ab.ca Phone: 1-866-421-6929

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